Singapore School Kinderland
Secondary-2
Computer Science (O
Level Standard: Sections
1.1 to 1.1.3)
Term 1

Class Test-1 (SECTIONS 1.1 TO 1.1.3 — ANSWER KEY & MARK SCHEME)

Date: _____ **Name: _____**

Total marks: 15 **Marks obtained: _____**

Teacher's signature: _____ **Parent's signature: _____**

Question 1: Data Representation & Multi-Base Conversions [Total: 4 marks]

O Level Exam Source: Cambridge 2210/12/M/J/24 & 2210/11/M/J/24

(a) State the role of logic circuits and transistors in why modern digital computers represent data using binary values. **[1 mark]**

Answer / Mark Scheme:
Computer microprocessors consist of millions of transistors that act as microscopic electronic switches with only two stable physical/electrical states: ON (voltage present / binary 1) and OFF (no voltage / binary 0) [1 mark].

(b) A 12-bit register contains the binary number:

1 1 0 0 1 0 1 0 0 1 1 1

Convert this 12-bit binary number into:
(i) Hexadecimal notation
(ii) Denary number (Show your calculation) **[3 marks]**

Answer / Mark Scheme:
(i) Hexadecimal: 1100 = C, 1010 = A, 0111 = 7 -> CA7 [1 mark]
(ii) Denary Calculation:
Method A (Place values): 2048 + 1024 + 128 + 32 + 4 + 2 + 1 = 3239
Method B (From Hex CA7): (12 * 256) + (10 * 16) + 7 = 3072 + 160 + 7 = 3239
[1 mark for calculation/working, 1 mark for correct denary value: 3239]

Question 2: Hexadecimal Conversions & Memory Storage [Total: 4 marks]

O Level Exam Source: Cambridge 2210/12/M/J/23 & 0478/12/M/J/23

(a) Convert the denary number 826 into Hexadecimal. Show your successive division working clearly. **[2 marks]**

Answer / Mark Scheme:
Working:
826 / 16 = 51 with remainder 10 (A in hex)
51 / 16 = 3 with remainder 3
3 / 16 = 0 with remainder 3
Reading remainders upwards / reverse: 33A [1 mark for method]
Hexadecimal: 33A [1 mark for final hex value]

(b) Convert the hexadecimal value 6E2 into binary format. **[1 mark]**

Answer / Mark Scheme:
Expand each hex digit to 4 bits: 6 = 0110, E (14) = 1110, 2 = 0010
Binary: 011011100010 [1 mark]

(c) State the minimum number of bytes required to store the hexadecimal number 1A4F in computer memory. Explain your answer. **[1 mark]**

Answer / Mark Scheme:
2 bytes. Each hexadecimal digit requires 4 bits (1 nibble). 4 hex digits = 16 bits = 2 bytes (since 1 byte = 8 bits) [1 mark].

Question 3: Media Access Control (MAC) & IP Addresses [Total: 4 marks]

O Level Exam Source: Cambridge 2210/12/M/J/24 & 2210/12/O/N/23

A network device has the following Media Access Control (MAC) address:

4 A : 5 B : 6 C : 7 D : 8 E : 9 F

(a) (i) Explain what the first 6 hexadecimal digits (4A:5B:6C) represent. **[1 mark]**

Answer / Mark Scheme:
The Manufacturer Identification Code / Organizationally Unique Identifier (OUI) [1 mark].

(a) (ii) Explain what the last 6 hexadecimal digits (7D:8E:9F) represent. **[1 mark]**

Answer / Mark Scheme:
The unique Device Serial Number assigned by the manufacturer [1 mark].

(b) Distinguish between a Universally Administered MAC Address (UAA) and a Locally Administered MAC Address (LAA). **[1 mark]**

Answer / Mark Scheme:
A UAA is set permanently by the manufacturer and cannot be changed, whereas an LAA can be configured/changed by the network administrator to bypass network restrictions [1 mark].

(c) IPv6 addresses are written using hexadecimal notation (e.g. 2001:0db8:85a3::8a2e:0370:7334). Explain why hexadecimal is used for IPv6 addresses instead of 128 binary bits. **[1 mark]**

Answer / Mark Scheme:
128 binary bits is too long and complex for human network engineers to read and configure;

grouping into 32 hex digits makes the address concise, manageable, and reduces configuration errors [1 mark].

Question 4: Memory Dumps & Error Diagnostics [Total: 3 marks]

O Level Exam Source: Cambridge 2210/11/O/N/23 & 2210/12/M/J/24

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During program debugging, an operating system outputs a Memory Dump
displaying RAM contents:

        Address: 0x00401A     Data: 4E  6F  74  20  46  6F  75  6E  64
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(a) Explain what is meant by a Memory Dump and describe how it helps software developers debug programs. **[2 marks]**

Answer / Mark Scheme:

A memory dump outputs the raw contents of RAM at the moment of a software crash in hexadecimal format [1 mark]. It allows programmers to examine memory locations, register states, and variable values to pinpoint the exact line or fault causing the crash [1 mark].

(b) State one other practical application of hexadecimal in computer systems (other than MAC addresses, IP addresses, and memory dumps). **[1 mark]**

Answer / Mark Scheme:

Any one from: Error codes (e.g. 0x80070057) / HTML-CSS RGB color codes (e.g. #FF5733) / Assembly language machine code representation [1 mark].